INTELLECTUAL PROPERTY DUE DILIGENCE REPORT

Project Lyra — proposed Patent and Know-How Licence from Veridian Photonics Limited to Marrick Defence Systems plc

Matter	Project Lyra (Whitfield Carter ref SW/JO/Lyra/00197)
Prepared by	Hartington IP LLP (patent attorneys to Veridian)
Prepared for	Whitfield Carter LLP / Veridian Photonics Limited
Date of report	2 May 2025
Scope	Confirmatory review of the VQ-1, VQ-2 and VQ-3 patent families and a freedom-to-operate screen within the def

1. Executive summary

We have reviewed the ownership, validity and status of the three patent families that Veridian Photonics Limited ("Veridian") proposes to license to Marrick Defence Systems plc ("Marrick") under Project Lyra, together with the associated know-how documentation. Subject to the specific matters in section 5, we consider the families to be properly owned by Veridian and the granted rights to be subsisting. The VQ-1 family (the core chip-scale sensor) is granted in GB, EP and the US and is the principal value driver. The VQ-2 family is at national-phase entry and the VQ-3 application has received a first examination report which, in our view, is answerable. No third-party right was identified that would prevent operation within the Field as the Licensed Technology is presently configured.

2. Ownership and chain of title

All three families are recorded at the relevant registries in the name of Veridian Photonics Limited (company number 11842207). The inventions originated with Veridian's founder-inventors (Dr Elaine Crawford and Dr Priyanka Rao) and were assigned to the company under employee IP assignments dated on incorporation; we reviewed executed copies of those assignments. There are no recorded security interests, licences of record or co-ownership entries against the VQ-1 or VQ-2 families. A non-commercial evaluation licence to the Cambridge Quantum Metrology Group is in place but is not registered and is limited to academic research.

3. Patent family status

Family	Representative right	Jurisdiction	Number	Status	Next action / rene
VQ-1 (QIS-Core)	Chip-scale cold-atom inertial sensor	GB	GB2598411	Granted	Renewal due Mar 2026
VQ-1 (QIS-Core)	(same)	EP	EP4012345	Granted (validated GB, FR, DE)	Annuities current
VQ-1 (QIS-Core)	(same)	US	US11,654,321	Granted	Maint. fee due 2027
VQ-2 (Atom-Trap)	MOT on photonic chip	PCT/GB	WO2024/118822	National phase (GB, EP, US)	EP/US filings due Aug 2025
VQ-3 (Bias-Cal)	Self-calibrating bias-drift method	GB	GB2611904	Pending (1st exam report)	Response due 8 Nov 2025

The VQ-1 family provides the broadest and most commercially significant protection and underpins the upfront consideration. The VQ-3 method claims are useful but secondary; the examiner's inventive-step objection over

reference D1 (US10,948,xxx) can in our view be overcome by amendment limiting the claims to the dynamically re-referenced calibration loop disclosed at paragraphs [0044]–[0051].

4. Freedom to operate (Field screen)

We conducted a targeted freedom-to-operate search within the defence/aerospace Field. One third-party European patent (EP3xxxxxx, expiring 2028) directed to a vapour-cell heating arrangement was identified as potentially relevant to the VQ-2 magneto-optical trap. Having reviewed the independent claims, we consider that the Licensed Technology as configured (which uses a grating-chip MOT without a separately heated vapour cell) does not fall within the scope of those claims. No other right of concern was identified. This screen is not a clearance opinion and should be refreshed before first commercial sale.

5. Specific points for the licence and disclosure

Ref	Point	Recommended treatment in the Licence
5.1	VQ-3 application not yet granted	Warranty 7.1(b) qualified; status disclosed in the Disclosure Letter (item 2.1).
5.2	Academic evaluation licence (CQM Group)	Disclosed (item 2.2); confirm it is non-commercial and field-restricted.
5.3	EP3xxxxxx FTO point on VQ-2	Disclosed (item 2.3); Licensor's awareness qualifier on warranty 7.1(d).
5.4	Export control of the technology	Addressed by clause 10; Marrick to confirm its own assessment.
5.5	Patent maintenance responsibility	Clause 6.1 places prosecution/maintenance cost on Veridian.

6. Know-how

The know-how to be transferred (document series VQ-KH-100 to VQ-KH-400) comprises fabrication specifications, control firmware, calibration datasets and qualification procedures. Veridian maintains these as trade secrets under documented access controls. We recommend the Licence require transfer into a secure data environment and that the confidentiality term run for at least seven years, with trade-secret material protected for so long as it remains secret (reflected in clause 9).

7. Conclusion

Subject to the specific points in section 5 being reflected in the Licence and Disclosure Letter, we consider the Licensed IP to be in a satisfactory condition to support the proposed exclusive, field-limited licence. We are content for Whitfield Carter LLP to rely on this report for the purposes of Project Lyra.

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This report is provided to Veridian Photonics Limited and its legal advisers in connection with Project Lyra and is subject to legal professional privilege. It is not a clearance or validity opinion and is given on the basis of the registers and documents available at the date of report.